

ELKHART-MORTON COUNTY AP, KS, USA

WMO: 724604

Lat: 37.000N

Lon: 101.883W

Elev: 3622

StdP: 12.87

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93076

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.2	12.3	-2.4	5.5	11.7	0.9	6.6	18.9	29.4	50.5	26.1	43.0	9.2	350	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.8	99.7	67.9	97.5	67.8	95.0	67.5	72.7	89.9	71.4	89.0	70.2	87.8	11.9	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	118.6	77.9	66.4	111.5	77.4	65.0	106.0	76.8	38.9	89.9	37.7	88.8	36.5	88.2	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	23.7	20.5	-0.2	104.0	3.5	2.0	-2.8	105.4	-4.8	106.6	-6.8	107.7	-9.4	109.1		
(5)			-0.7	74.0	4.1	1.8	-3.6	75.3	-6.1	76.4	-8.4	77.4	-11.3	78.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	36.3	38.5	47.4	54.9	64.7	75.6	79.4	77.4	70.4	57.6	46.5	36.8	(6)
(7)		DBStd	17.92	10.53	11.59	10.51	9.69	8.77	6.36	5.11	5.56	7.29	9.09	9.80	10.47	(7)
(8)		HDD50	1649	434	346	172	60	6	0	0	0	35	177	419		(8)
(9)		HDD65	4350	890	742	548	321	114	6	1	2	34	260	557	875	(9)
(10)		CDD50	4287	9	24	91	207	463	768	912	850	613	270	71	9	(10)
(11)		CDD65	1510	0	0	2	18	106	323	448	386	196	30	1	0	(11)
(12)		CDH74	18501	4	19	131	505	1571	3980	5324	4166	2202	539	59	1	(12)
(13)		CDH80	9656	0	3	23	158	709	2184	3063	2275	1048	188	5	0	(13)
(14)	Wind	WSAvg	10.6	9.8	10.4	11.2	12.5	11.7	11.4	9.8	9.4	10.4	10.3	10.3	9.8	(14)
(15)	Precipitation	PrecAvg	17.5	0.4	0.6	1.2	1.5	2.5	2.5	2.7	2.2	1.9	0.9	0.8	0.4	(15)
(16)		PrecMax	29.2	1.9	1.8	4.0	5.4	8.3	5.4	6.2	7.6	5.3	3.7	3.7	1.3	(16)
(17)		PrecMin	8.8	0.0	0.0	0.0	0.0	0.2	0.2	0.4	0.3	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.8	0.4	0.6	1.1	1.5	1.6	1.5	1.7	1.7	1.4	0.9	0.9	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.7	76.9	82.7	89.9	97.1	102.6	102.2	101.1	98.3	90.4	80.0	70.6	(19)
(20)			MCWB	48.3	51.0	54.6	57.7	62.0	65.7	69.1	69.4	66.3	60.6	53.9	48.6	(20)
(21)		2%	DB	64.5	70.1	78.0	84.3	91.9	99.0	99.7	98.7	93.9	85.6	74.6	63.7	(21)
(22)			MCWB	44.9	48.2	52.7	55.5	61.5	66.5	68.7	68.7	65.9	59.3	51.5	45.6	(22)
(23)		5%	DB	58.5	64.1	72.6	80.7	88.0	96.0	97.5	95.5	90.4	80.7	69.2	58.4	(23)
(24)			MCWB	42.6	45.3	50.5	54.6	60.5	66.4	68.7	68.9	65.6	57.7	49.3	42.8	(24)
(25)		10%	DB	53.0	57.3	66.6	74.8	83.3	92.3	94.9	92.5	86.5	74.8	63.1	53.2	(25)
(26)			MCWB	39.7	42.4	48.0	52.9	60.0	66.1	68.8	68.2	64.7	55.5	46.7	40.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.6	51.4	58.4	61.1	69.1	73.1	74.8	74.0	71.7	67.8	55.9	49.5	(27)
(28)			MCD B	68.9	73.8	76.3	78.4	82.7	90.6	92.4	90.2	87.4	76.1	76.2	68.5	(28)
(29)		2%	WB	45.9	48.7	54.5	58.4	66.2	71.1	73.0	72.4	69.6	62.7	52.6	46.4	(29)
(30)			MCD B	62.9	68.1	71.6	77.3	81.9	89.3	90.8	89.0	85.7	76.8	71.0	62.4	(30)
(31)		5%	WB	43.1	45.8	51.6	56.3	63.8	69.4	71.6	71.1	68.0	59.7	50.3	43.6	(31)
(32)			MCD B	58.3	62.8	70.0	74.7	80.2	87.7	89.7	88.3	84.4	75.9	66.6	57.6	(32)
(33)		10%	WB	40.1	42.6	49.1	54.3	61.7	68.0	70.2	69.8	66.2	57.1	47.7	40.7	(33)
(34)			MCD B	52.2	57.2	65.9	71.9	78.6	86.3	88.7	86.8	81.5	73.6	62.5	52.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	24.9	26.5	27.7	27.4	27.4	27.8	26.8	25.5	26.5	26.7	26.0	23.2	(35)
(36)			MCDBR	33.5	35.7	35.4	36.2	34.0	32.7	30.6	30.4	31.2	34.1	34.2	31.2	(36)
(37)			MCWBR	19.5	19.8	17.6	16.4	13.7	11.2	10.3	9.9	11.4	15.4	17.7	18.0	(37)
(38)		5% WB	MCD BR	31.9	34.4	32.9	31.6	30.0	28.9	27.8	26.9	27.2	29.3	31.9	30.2	(38)
(39)			MCWBR	19.0	19.6	17.2	15.4	13.3	11.5	10.3	10.2	10.6	14.6	17.2	18.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.254	0.261	0.286	0.313	0.335	0.367	0.369	0.356	0.331	0.296	0.268	0.252	(40)	
(41)		taud	2.581	2.569	2.505	2.442	2.440	2.383	2.428	2.432	2.485	2.553	2.563	2.588	(41)	
(42)		Ebn at Noon	305	314	313	307	300	289	287	290	293	296	296	299	(42)	
(43)		Edh at Noon	23	27	31	35	36	38	36	35	32	27	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	902	1165	1577	1950	2241	2407	2351	2070	1756	1316	991	793	(44)	
(45)		RadStd	76	108	132	106	128	111	95	102	87	133	81	51	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	+1.40	+2.34	N/A	N/A	N/A	N/A	
(47) Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+130	N/A	

Nomenclature: See separate page